Basic Course on Foundation of Cryptography, Lecture 12 Black-Box Impossibilities[1]

Handout Mode

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Talk plan

- Motivation and definition
- Random permutations are hard to invert
- Impossibility for Basing OWP on OWF
- Impossibility for basing efficient PRG on OWF

Motivating example: Basing Key-Agreement on OWFs

- Key-Agreement protocols (KA) can be based on the existence of TDP, RSA or discrete log assumptions, and ...
- We don't know how to base KA on the existence of OWFs/OWPs.

- Can we base KA on OWFs/OWPs?
- Proving unconditional negative result seems beyond reach.

  Assume RSA assumption holds.

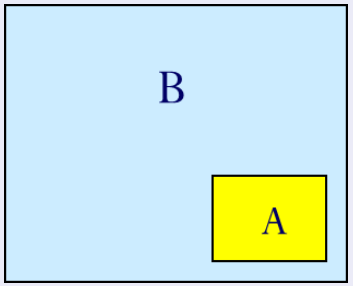
  $\Longrightarrow$ key-agreement protocols exist.

  $\Longrightarrow$ OWFs imply the existence of key-agreement protocols in a trivial sense.

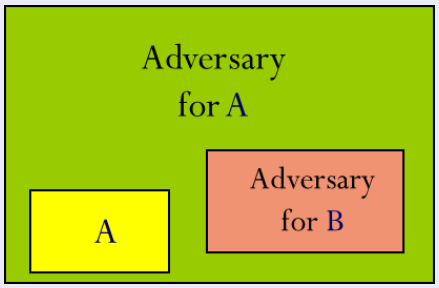
(Fully) Black-box constructions

Definition 1 (A fully Black-box construction of B from A)

Black-box **construction**: Oracle-aided PPT C such that C^f implements B for any algorithm f implementing A.



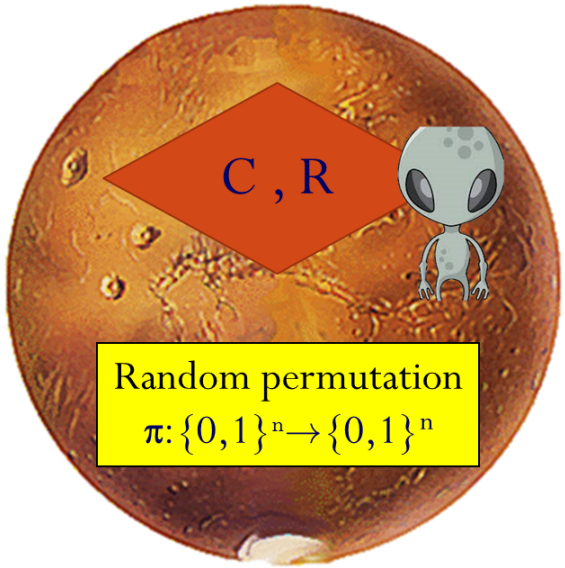
Black-box **proof of security**: Oracle-aided PPT R such that $\mathrm{R}^{f,\mathrm{Brk}}$ breaks f, for any algorithms f implementing A, and Brk breaking C^f.



- Fully-black-box constructions relativize: hold relative to any oracle.
- Most constructions in cryptography are (fully) black-box, e.g., pseudorandom generator from OWF.
- Some “non black-box" techniques apply in restricted settings

Proving BB impossibility result

Assume $\exists$ fully-BB construction (C, R) of a KA from OWP.



Section 1

Random Permutations are Hard To invert

Some preliminaries

- $k! := 1 \cdot 2 \cdot 3 \cdot \cdots \cdot k$
- $\binom{k}{a} := \frac{k!}{a! \cdot (k-a)!}$
- $k! = \binom{k}{a} \cdot a! \cdot (k-a)!$
- $k! \geq (\frac{k}{e})^k$ and $\binom{k}{a} \leq (\frac{ek}{a})^a$

Let $N = 2^n$.

- How many bits it takes to describe $\mathcal{S} \subseteq \{0,1\}^n$ of (known) size a? There are $\binom{N}{a}$ such sets, so it takes $\log \binom{N}{a}$ bits.

Let Π_n be the set of all permutations over $\{0,1\}^n$.

Q1: How many bits it takes to describe $\pi \in \Pi_n$?

A1: $|\Pi_n| = N!$, so it takes $\log(N!)$ bits.

Q2: and to describe invective and onto function from $\mathcal{D}$ to $\mathcal{R}$?

A2: $\log(|\mathcal{D}|!)$ (note that $|\mathcal{D}| = |\mathcal{R}|$)

Some preliminaries, cont.

What is the number of of M-size oracle-circuits?

Claim 2 (HW 1)

Number of M-size oracle-circuits mapping n-bit strings to n-bit strings, with oracle access to a function n-bit strings to n-bit strings, is at most $2^{2M+(M+1)n(\log(Mn+n)+1)}$.

Notation

For circuit (or deterministic algorithm) D and $f\colon \{0,1\}^n \mapsto \{0,1\}^n$, let

- $\mathrm{Inv}_f^{\mathsf{D}}(y) = \begin{cases} 1, & \mathsf{D}(y) \in f^{-1}(y), \\ 0, & \text{ow.} \end{cases}$
- $\mathrm{Inv}^{\mathsf{D}}(f) = \mathrm{E}_{x \leftarrow \{0,1\}^n}\left[\mathrm{Inv}_f^{\mathsf{D}}(f(x))\right]$.

Random permutations are hard to invert

Theorem 3 (Gennarro-Tevisan, '01)

For large enough $n \in \mathbb{N}$ and $2^{n/5}$-query circuit D,

$$\Pr_{\pi \leftarrow \Pi_n}\left[\mathsf{Inv}^{\mathsf{D}}(\pi) = \Pr_{y \leftarrow \{0,1\}^n}\left[\mathsf{D}(y) = \pi^{-1}(y)\right] > 2^{-n/5}\right] \leq 2^{-2^{3n/5}}$$

Random permutations are exponentially hard for exponential-query circuits.

- Extends to randomized algorithms.
- Constants are somewhat arbitrary and non tight.
- Since $\#$ of $2^{n/5}$-size circuits is bounded by $2^{\tilde{O}(2^{n/5})}$, Thm 3 yields that

$$\Pr_{\pi \leftarrow \Pi_n}\left[\exists\ 2^{n/5}\text{-size circuit D with } \mathsf{Inv}^{\mathsf{D}}(\pi) > 2^{-n/5}\right] \leq 2^{-2^{n/2}}$$

Random permutations are exponentially hard for all exponential-size circuits simultaneously.

Proving GT theorem (Thm 3), lazy evaluation approach

- Lazy evaluation approach: for any $2^{n/5}$-size circuit C and $x \in \{0,1\}^n$

$$\Pr_{\pi \leftarrow \Pi_n}[C^\pi(\pi(x)) = x] \leq 2^{-4n/5}$$

- It follows that for any $2^{n/5}$-size circuit C

$$\Pr_{\pi \leftarrow \Pi_n}\left[\Pr_{x \leftarrow \{0,1\}^n}[C^\pi(\pi(x)) = x] > 2^{-3n/5}\right] < 2^{-n/5}$$

- Not strong enough to rule our inversion by all small circuits simultaneously.

Proving GT theorem (Thm 3), warmup

Assume D makes no queries.

For how many $\pi \in \Pi_n$ it holds that $\mathsf{Inv}^{\mathsf{D}}(\pi) = 1$?

Answer: one! (defined by $\pi^{-1}(y) = \mathsf{D}(y)$)

Hence,

$$\Pr_{\pi \leftarrow \Pi_n}\left[\mathsf{Inv}^{\mathsf{D}}(\pi) = 1\right] \leq \frac{1}{N!} \leq \left(\frac{e}{2^n}\right)^{2^n}$$

For how many $\pi \in \Pi_n$ it holds that $\mathsf{Inv}^{\mathsf{D}}(\pi) \geq \varepsilon$?

What if D makes oracle queries?

Proving GT theorem (Thm 3)

Lemma 4 (Compression lemma)

For any q-query circuit D and $\varepsilon > 0$, exist Enc and Dec s.t. the following holds: Let $\pi \in \Pi_n$ be s.t. $\mathsf{Inv}^{\mathsf{D}}(\pi) > \varepsilon$, then

- $\mathsf{Dec}(\mathsf{Enc}(\pi)) = \pi$.
- $|\mathsf{Enc}(\pi)| \leq \log((N-a)!) + 2 \cdot \log \binom{N}{a}$, *for some* $a \geq \frac{\varepsilon N}{q+1}$.

$\implies$ $\#$ of permutations in Π_n with $\mathsf{Inv}^{\mathsf{D}}(\pi) > \varepsilon$ is at most $(N-a)! \cdot \binom{N}{a}^2$.

- Let D be a $q = 2^{n/5}$-query circuit. By Lemma 4

$$\Pr_{\pi \leftarrow \Pi_n}\left[\mathsf{Inv}^{\mathsf{D}^\pi}(\pi) > 2^{-n/5}\right] \leq \frac{(N - 2^{\frac{3}{5}n})! \cdot \binom{N}{2^{\frac{3}{5}n}}^2}{N!} = \frac{\binom{N}{2^{\frac{3}{5}n}}}{2^{\frac{3}{5}n}!} \leq 2^{-2^{3n/5}}$$

proving Thm 3.

Proving Compression Lemma, warm-up

Assume D makes no queries.

- Fix $\pi \in \Pi_n$ with $\mathsf{Inv}^{\mathsf{D}}(\pi) > \varepsilon$, and let $\mathcal{Y}_\pi = \{y \in \{0,1\}^n \colon \mathsf{D}(y) = \pi^{-1}(y)\}$

  Note that $|\mathcal{Y}| \geq \varepsilon N$.

- π is determined by $\mathcal{Y}_\pi$ and $\mathcal{V}_\pi = \{(x, \pi(x)) \colon \pi(x) \notin \mathcal{Y}_\pi\}$.

  Proof: π is determined by the sets $\mathcal{V}_\pi$ and $\mathcal{V}'_\pi = \{(\mathsf{D}(y), y) \colon y \in \mathcal{Y}_\pi\} = \{(x, \pi(x)) \colon \pi(x) \in \mathcal{Y}_\pi\}$.

- Given $\mathcal{Y}_\pi$, the set $\mathcal{V}_\pi$ is determined by the set $\mathcal{X}_\pi = \pi^{-1}(\mathcal{Y}_\pi) = \{\pi^{-1}(y) \colon y \in \mathcal{Y}_\pi\}$, and a permutation $\tau_{\mathcal{Y}}$ over $\{0,1\}^n \setminus \mathcal{X}_\pi$.

- Hence, see Q2, π can be described using $2 \cdot \log \binom{N}{|\mathcal{Y}|} + \log((N - |\mathcal{Y}|)!)$ bits.

Define $\mathsf{Enc}(\pi)$ to output the description of $\mathcal{Y}_\pi$, and $\mathcal{V}_\pi$.

$\mathsf{Dec}(\mathcal{Y}, \mathcal{X}, \tau)$ is defined accordingly.

Proving Compression Lemma

Let D be a q-query circuit, and let π be s.t. $\mathsf{Inv}^{\mathsf{D}^\pi}(\pi) > \varepsilon$.

Construction 5 (Useful set $\mathcal{Y}_\pi \subseteq \{0,1\}^n$)

1. Set $\mathcal{Y}_\pi = \emptyset$ and $\mathcal{I}_\pi = \{y \in \{0,1\}^n \colon \mathsf{D}^\pi(y) = \pi^{-1}(y)\}$.
2. While $\mathcal{I} \neq \emptyset$, let y be the smallest lexicographic element in $\mathcal{I}_\pi$.
   (a) Add y to $\mathcal{Y}_\pi$.
   (b) Remove y and all answers to π-queries $\mathsf{D}^\pi(y)$ makes, from $\mathcal{I}_\pi$.

Algorithm 6 ($\mathsf{Enc}(\pi)$)

Output (description of) $\mathcal{Y}_\pi$ and $\mathcal{V}_\pi = \{(x, \pi(x)) \colon \pi(x) \notin \mathcal{Y}_\pi\}$.

(Under proper encoding) $|\mathsf{Enc}(\pi)| \leq \log((N-a)!) + 2 \cdot \log \binom{N}{a}$ for $a = |\mathcal{Y}_\pi| \geq \frac{\varepsilon N}{q+1}$.

Algorithm 7 ($\mathsf{Dec}(\mathcal{Y}_\pi, \mathcal{V}_\pi)$)

For all $y \in \mathcal{Y}_\pi$ in lexicographic order:

1. Emulate $\mathsf{D}^\pi(y)$, answering π-query using $\mathcal{V}_\pi$.
2. If D queries x that is undefined in $\mathcal{V}$, add (x, y) to $\mathcal{V}_\pi$.
   Otherwise, add $(\mathsf{D}^\pi(y), y)$ to $\mathcal{V}_\pi$.

Output $\mathcal{V}_\pi$.

Remarks

- Result can be proven using alternative compression arguments.
- Extends to random functions and random trapdoor permutation families (TDPs).

  HW2: state and prove for random function from n bit to n bits

- The short description argument is an very useful paradigm (another example soon).

Section 2

BB Impossibility for Basing OWP on OWF

OWF vs OWP

- Most of what we can based on OWPs can also be based on OWFs

  But constructions are much less efficient and way more complicated
- Can we base OWF on OWP?
- Actually the “correct” question, which we do not know how to answer, is: Can we base “almost” invective OWF on OWF?
- We will show that we we cannot base OWP on OWF in a (fully) black box way

The settings

- Let g be an $q(n)$-query oracle-aided function such that $g^f\colon \{0,1\}^{m(n)} \mapsto \{0,1\}^{m(n)}$ is a permutation for every $f\colon \{0,1\}^n \mapsto \{0,1\}^n$.
- For a start, assume that m is increasing and efficiently invertible, and that on input of length $m(n)$, g only call f on inputs of length n
- R be the reduction: R is an efficient algorithm such that $\mathsf{R}^{f,\mathsf{Inv}}$ breaks the one-wayness of $f\colon \{0,1\}^n \mapsto \{0,1\}^n$ for any Inv that breaks that of g^f.
- For a start, assume for simplicity that on input of length n, R only calls f on input of length n

The attack

Algorithm 8 (The inverter $\mathsf{Inv}^f(z \in \{0,1\}^{m=m(n)})$)

For $i = 1$ to $q = q(n)$

- Find (brute forcly) $x \in \{0,1\}^m$ and $f' \colon \{0,1\}^n \mapsto \{0,1\}^n$ consistent with $\mathcal{S}$ (initially empty) such that $g^{f'}(x) = z$
- Add $(x_1, f(x_1)), \ldots, (x_q, f(x_q))$ to $\mathcal{S}$, for $x_1, \ldots, x_q$ being the queries made by $g^{f'}(x)$

Return $x \in \{0,1\}^m$ such that all queries of $g^f(x)$ are in $\mathcal{S}$ and $g^f(x) = z$.

- On $z \in \{0,1\}^{m(n)}$, Inv makes at most $q(n)^2$ queries.
- Inv^f inverts g^f with probability one. Proof
  - Let $x = (g^f)^{-1}(z)$.
  - In each round, $\mathsf{Inv}(z)$ asks at least one additional query of $g^f(x)$ (unless it already found all of them) (?)
  - When the loop ends, $\mathsf{Inv}(z)$ asked all queries made by $g^f(x)$, and hence finds x

The impossibility result

- By assumption, for any $f\colon \{0,1\}^n \mapsto \{0,1\}^n$, $\exists p \in \text{poly}$ such that

$$\Pr_{x \leftarrow \{0,1\}^n, y=f(x)} \left[\mathsf{R}^{\mathsf{Inv},f}(y) \in f^{-1}(y) \right] \geq 1/p(n)$$

- $\mathsf{R}^{\mathsf{Inv},f}(y \in \{0,1\}^n)$ makes polynomial number of queries.
- For large enough n, let $f_n\colon \{0,1\}^n \mapsto \{0,1\}^n$ be such that no $2^{n/5}$-query algorithm inverts with probability larger than $2^{-n/5}$, and let $f = \{f_n\}$
- Hence, $\Pr_{x \leftarrow \{0,1\}^n, y=f(x)} \left[\mathsf{R}^{\mathsf{Inv},f}(y) \in f^{-1}(y) \right] < 2^{-n/5}$ for every large enough n.

  A contradiction.
- What goes wrong if we allow g to query f on different input length?
- Essentially nothing, HW 3
- What goes wrong if we allow R to query f on different input lengths?

Handling reductions of arbitrary query lengths

- Let $\mathcal{F}_n$ be all length-preserving functions over $\{0,1\}^n$ and let $\mathcal{F} = \{\mathcal{F}_n\}_{n\in\mathbb{N}}$
- For large enough n,

  $\Pr_{\pi\leftarrow\mathcal{F}}\left[\Pr_{x\leftarrow\{0,1\}^n, y=f(x)}\left[\mathsf{R}^{\mathsf{Inv},f}(y) \in f^{-1}(y)\right] \geq 2^{-n/5}\right] < 2^{-2^{3n/5}}$

Lemma 9 (Borel-Cantelli)

Let $E_1, \ldots, E_2, \ldots,$ be a sequence of events such that $\sum_{n=1}^{\infty} \Pr[E_n] < \infty$, then $\Pr\left[\limsup_{n\to\infty} E_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k\right] = 0$.

Namely, with probability one, only finite number of event occur.

- Let E_n be the event over the choice of $f \leftarrow \mathcal{F}$ that $\Pr_{x\leftarrow\{0,1\}^n, y=f(x)}\left[\mathsf{R}^{\mathsf{Inv},f}(y) \in f^{-1}(y)\right] \geq 2^{-n/5}$
- (for large enough n) $\Pr\left[E_n\right] \leq 2^{-2^{3n/5}} < 1/n^2$
- Hence, $\sum_{n=1}^{\infty} \Pr[E_n] \leq O(1) + \sum_{n=1}^{\infty} 1/n^2 \in O(1)$
- Hence, with probability one over the choice of f, only for finite number of n it holds that $\Pr_{x\leftarrow\{0,1\}^n, y=f(x)}\left[\mathsf{R}^{\mathsf{Inv},f}(y) \in f^{-1}(y)\right] \geq 2^{-n/5}$.

Remarks

- Extend to many constructions of prefect object from a imperfect object
- Other example: Perfect KA from OWF, perfect encryption schemes from a non perfect one.
- Seem to be “less significant” impossibility results
- In some cases, and under the right assumptions, can be bypassed using non-black-box technique

  (next class)

Section 3

BB Impossibility for Efficient OWF based PRG

BB Impossibility for OWF based PRG

Definition 10 (Pseudorandom generators (PRGs))

Poly-time $G\colon \{0,1\}^n \mapsto \{0,1\}^{\ell(n)}$ is a pseudorandom generator, if

- G is length extending (i.e., $\ell(n) > n$ for any n)
- $G(U_n)$ is pseudorandom (i.e., $\{G(U_n)\}_{n\in\mathbb{N}} \approx_c \{U_{\ell(n)}\}_{n\in\mathbb{N}}$)

We focus on BB constructions of efficient length-doubling PRGs.

Theorem 11

In any fully-BB construction of length-doubling PRG over n-bits string from OWP over $\{0,1\}^n$, the construction makes $\Omega(n/\log n)$ oracle calls.

- Matches known upper bounds.
- Without the restriction on the OWP input length, yields an optimal $n^{\Omega(1)}$ bound.

Proving Thm 11, cont.

- Let (C, R) be a fully-BB reduction of a $q(n) \in o(n/\log n)$-query, length-doubling PRG over $\{0,1\}^n$, to OWP over $\{0,1\}^n$.
- Assume wlg. that C makes distinct queries.
- Assume for simplicity that on inputs of length n, R only makes lenght n queries.
- For $t = t(n) = \lfloor n/2q(n) \rfloor$, consider the following generator $G\colon \{0,1\}^{\frac{3}{2}n} \mapsto \{0,1\}^{2n}$:

Algorithm 12 ($G(x)$)

1. Emulate $\mathrm{C}^{\pi}(x_{1,\ldots,n})$, while answering the i'th query z of C to π, with $x_{n+i\cdot t+1,\ldots,n+(i+1)\cdot t} \circ z_{t+1,\ldots,n}$. (copy the last $n-t$ bits)
2. Output the same output that C does.

- Let $\Pi_{n,t}$ be the set of all permutations over $\{0,1\}^n$ that are identity over the last $n-t$ bits (i.e., $\pi(x)_{n-t+1,\ldots,n} = x_{n-t+1,\ldots,n}$).
- $\mathrm{SD}\left(G(U_{3n/2}), (\mathrm{C}^{\pi}(U_n))_{\pi \leftarrow \Pi_{n,t}}\right) \leq q^2/2^t < 1/4$ (?)

Proving Thm 11, cont.

- $\exists$ algorithm D that distinguishes $G(U_{3n/2})$ from U_{2n} with advantage $1 - 2^{-n/4} > \frac{3}{4}$. (?)

  $\implies$ wlg. $\Pr_{\pi \leftarrow \Pi_{t,n}}[\mathrm{D}(\mathrm{C}^{\pi}(U_n)) = 1] - \Pr[\mathrm{D}(U_{2n}) = 1] > \frac{1}{2}$

  $\implies \Pr_{\pi \leftarrow \Pi_{t,n}}\left[\Pr[\mathrm{D}(\mathrm{C}^{\pi}(U_n)) = 1] - \Pr[\mathrm{D}(U_{2n}) = 1] > \frac{1}{4}\right] \geq \frac{1}{4}$

  $\implies \Pr_{\pi \leftarrow \Pi_{t,n}}\left[\mathrm{R}^{\pi,\mathrm{D}} \text{ inverts } \pi \text{ with non-negligible prob.}\right] \geq \frac{1}{4}$
- Let $n' = t(n) \in \omega(\log n)$.
- Sine R makes $\mathrm{poly}(n)$ queries, exists $2^{o(n')}$-query circuit R', such that

$$\Pr_{\pi \leftarrow \Pi_{n'}}[\mathrm{R}'^{\pi} \text{ inverts } \pi \text{ with non-negligible prob.}] \geq \frac{1}{4},$$

in contradiction to Thm 3.

Remarks

- We showed a lower bound on the efficiency of fully-BB constructions of length-doubling PRG from OWPs.
- Results extend to OWFs and TDPs.
- Using similar means, one can prove lower bound on BB constructions of encryption schemes, signature schemes and universal-one-way-hash-functions (UOWHFs), from OWFs/OWPs/TDPs
- The attacker does not use the oracle to f
- Result can be extended to rule out a **less restricted** type of BB-construction, called weak black box:

  If f is a secure implementation of A, then C^f is a secure implementation of B.